

Abstract Algebra II

Homework 1

Due: 11th March 2026

Submit only Questions 1 and 2.

1. Let K be a field and L an extension of K . For a subset of elements $\{u_1, \dots, u_r\} \subset L$, we denote by $K(u_1, \dots, u_r)$ the field consisting of elements of the form

$$\frac{g(u_1, \dots, u_r)}{h(u_1, \dots, u_r)}, \quad g, h \in K[x_1, \dots, x_r], \quad h \neq 0.$$

This is the smallest subfield of L containing K and $u_1, \dots, u_r$. Let $u \in L$ be algebraic over K . Prove that

- (a) $K(u) \cong K[u]$;
 - (b) $K(u) \cong K[x]/(f)$, where $f \in K[x]$ is an irreducible monic polynomial of degree $n \geq 1$ satisfying the following conditions
 - i) $f(u) = 0$;
 - ii) for $g \in K[x]$, $g(u) = 0$ if and only if f divides g .
2. Let L and L' be field extensions of a field K and $u \in L$, $u' \in L'$ be algebraic over K . If there is an isomorphism of fields $K(u) \cong K(u')$ mapping u to u' , then u and u' are roots of the same irreducible polynomial $f \in K[x]$.
 3. If K is a field and $f \in K[x]$ is a polynomial of degree n , then there exists a simple extension field $F = K(u)$ of K such that:
 - (a) $u \in F$ is a root of f ;
 - (b) $[K(u) : K] \leq n$, with equality holding if and only if f is irreducible.
 4. Let F/K be a field extension. If $[F : K]$ is finite, then this field extension is algebraic.